CS & IT ENGINEERING

Operating System

Virtual Memory

Lecture - 3

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Recap of Previous Lecture

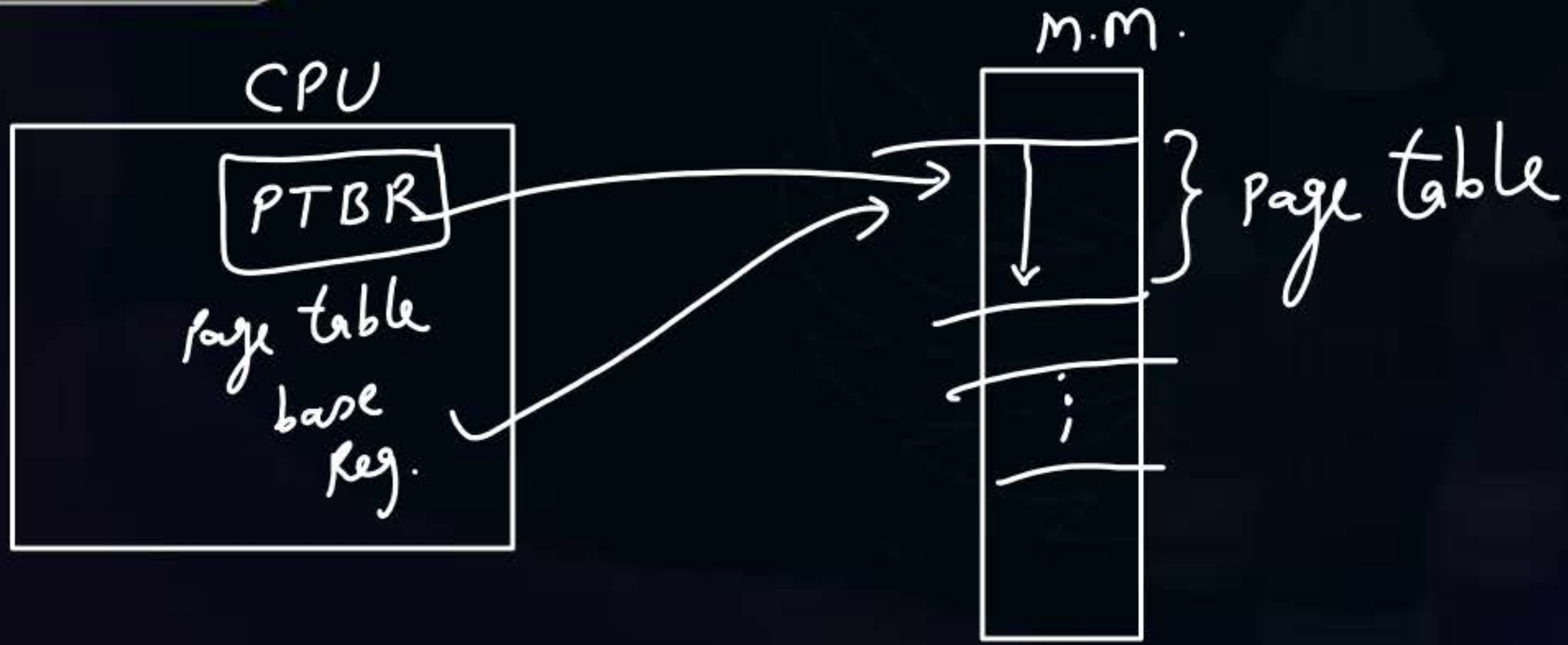
Topic Effective Memory Access Time

Topic Page Replacement

Topics to be Covered

Topic Page Replacement

Topic FIFO, Optimal, LRU Policies



Topic : Question

154.5 to 155.5

[GATE-2020]

#Q. Consider a paging system that uses 1-level page table residing in main memory and a TLB for address translation. Each main memory access takes 100 ns and TLB lookup takes 20 ns. Each page transfer to/from the disk takes 5000 ns. Assume that the TLB hit ratio is 95%, page fault rate is 10%. Assume that for 20% of the total page faults, a dirty page has to be written back to disk before the required page is read from disk. TLB update time is negligible. The average memory access time in ns (round off to 1 decimal places) is________.

$$= 20 + (0.95 * 100) + 0.05\left[100 + 0.9 * 100 + 0.1 * \left(0.8 * (5000 + 100) + 0.2 * (5000 + 5000 + 100)\right)\right]$$

$$= 155 \text{ ns}$$

virtual memory with cache & TLB :- (no page fault)

physically addressed cache

avg content access time with physical address $= H_c * t_{cm} + (1 - H_c) * (t_{cm} + t_{mm})$
(t_{act})

Where,

H_c = hit rate of cache

t_{cm} = cache access time

t_{mm} = main mem. —||—

CPU generates

⇓

Logical address

⇓

search in TLB

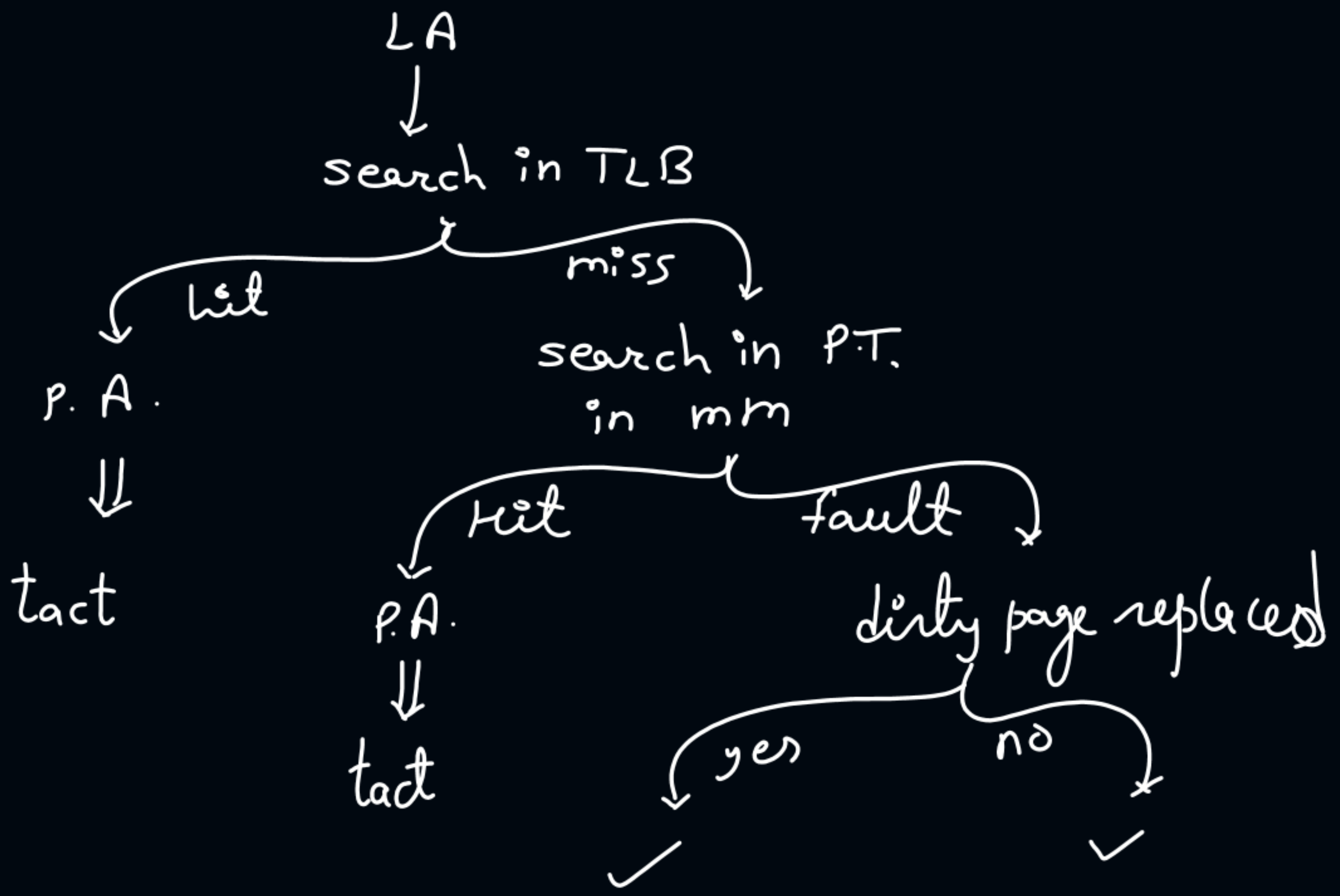
TLB hit → P.A. ⇓ search in Cache

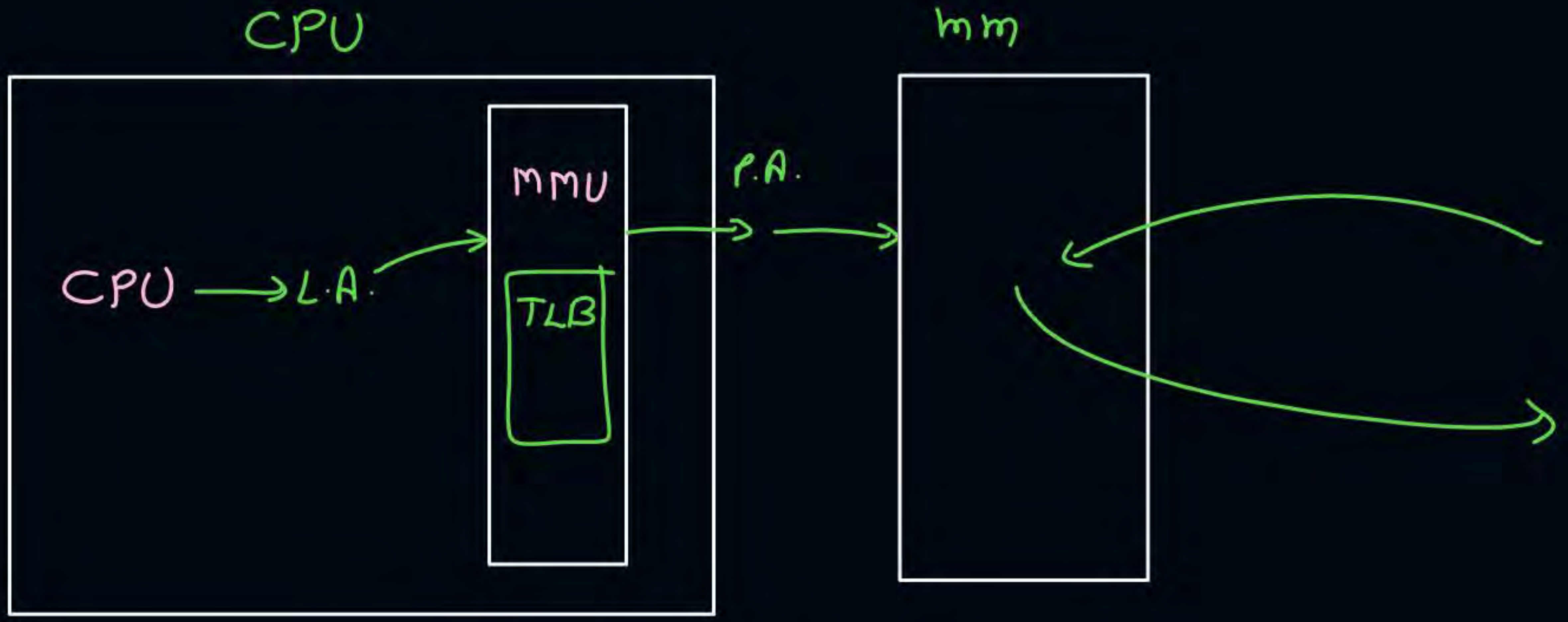
- hit → access content from cache
- miss → access content from mm

TLB miss → search P.T. in mm ↓ P.A. ↓ search in Cache

- hit → access content from cache
- miss → access content from mm

$$E.M.A.T. = t_{TLB} + H * t_{act} + (1-H)\left[t_{mm} + t_{act}\right]$$


LA
search in TLB
hit
miss
P. A.
tact
search in P.T.
in mm
Hit
fault
P.A.
tact
dirty page replaced
yes
no
✓
✓



MMU : Memory Management Unit

Topic : Page Replacement Policies

1. First In First Out (FIFO)
2. Optimal Policy
3. Least Recently Used (LRU)
4. Least Frequently Used (LFU)
5. Most Frequently Used (MFU)
6. Last In First Out (LIFO)
7. Second Chance

Topic : First In First Out (FIFO)

replace a page which comes first in main memory

Assume:

- Number of frames = 3 (All empty initially)
- Page reference sequence: 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5

1	2	3	4	1	2	5	1	2	3	4	5
1	1	1	4	4	4	5	5	5	5	5	5
	2	2	2	1	1	1	1	1	3	3	3
		3	3	3	2	2	2	2	2	4	4
✓	✓	✓	✓	✓	✓	✓	Hit	Hit	✓	✓	Hit

No. of page faults = 9

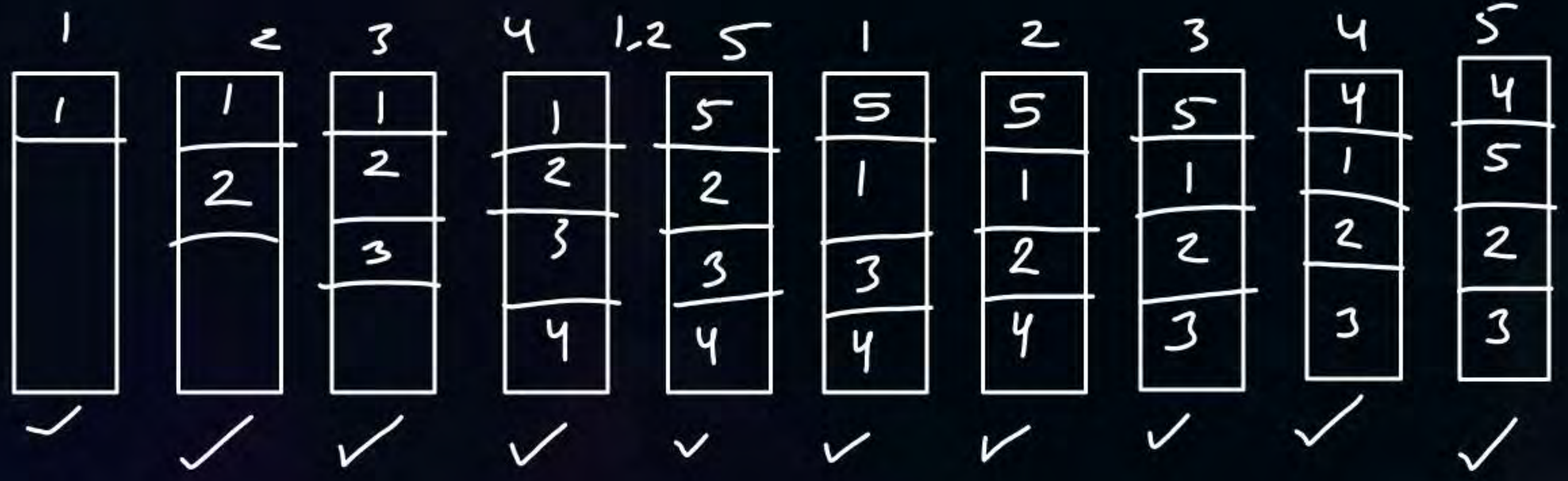
No. of page hits = 3

Page fault rate = $\frac{9}{12}$ | Page hit rate = $\frac{3}{12}$

Topic : First In First Out (FIFO)

Assume:

- Number of frames = 4 (All empty initially)
- Page reference sequence: 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5



no. of page faults = 10

Topic : Belady's Anomaly

Increasing the no. of frames can increase no. of page faults for some page reference sequences.

$\Downarrow$

only FIFO suffer's from belady's anomaly

Topic : First In First Out (FIFO)

Advantages:

1. Simple and easy to implement.
2. Low overhead. ⇒ for searching which page to replace

Disadvantages:

1. Poor performance. ⇒ High page fault rate
2. Doesn't consider the frequency of use or last used time, simply replaces the oldest page.
3. Suffers from Be lady's Anomaly

Topic : Optimal Policy

replace a page which will not be used for longest period of time in future

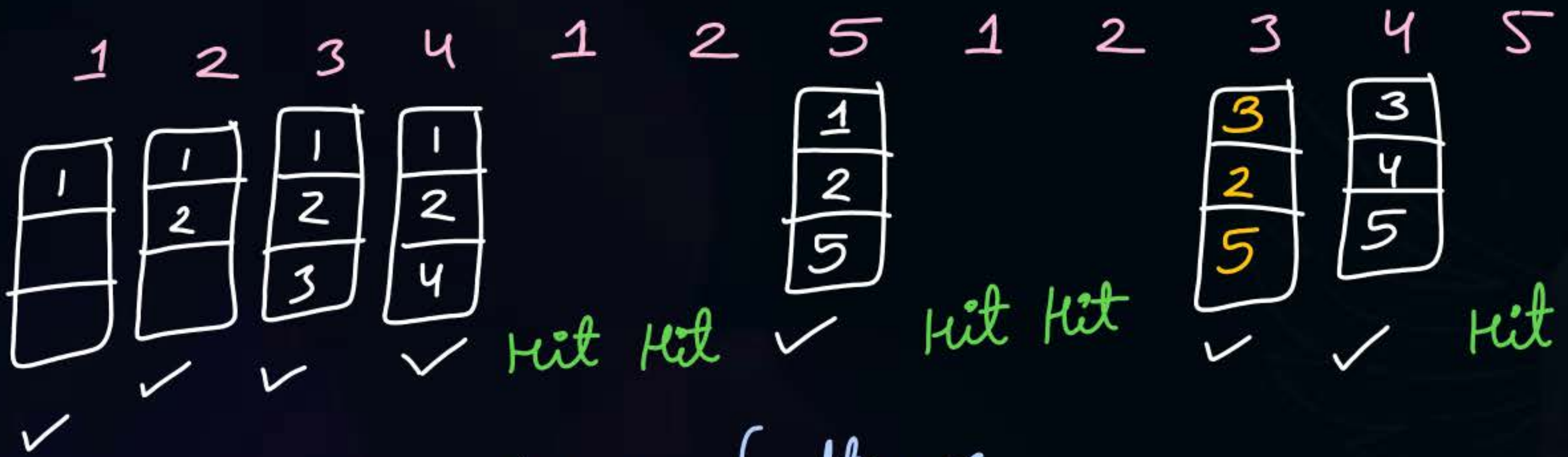
⇓

Tie breaker

↳ Use FIFO

Assume:

- Number of frames = 3 (All empty initially)
- Page reference sequence: 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5



no. of page faults = 7

Topic : Optimal Policy

Advantages:

1. Easy to Implement
2. Simple data structures are used
3. Highly efficient as it provides min. no. of page faults

Disadvantages:

1. Requires future knowledge of the program hence practically can't be implemented
2. Time-consuming

↳ To search which page to replace.

no. of page fault using optimal policy $\leq$ no. of page faults using any page replacement policy

Topic : Least Recently Used (LRU)

replace a page which has not been used for longest period of time.

Assume:

- Number of frames = 3 (All empty initially)
- Page reference sequence: 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5

1	2	3	4	1	2	5	1	2	3	4	5
1	1	1	4	4	4	5			3	3	3
	2	2	2	1	1	1			1	4	4
		3	3	3	2	2			2	2	5
✓	✓	✓	✓	✓	✓	✓	hit	hit	✓	✓	✓

no. of page faults = 10

Topic : Least Recently Used (LRU)

Advantages:

1. Efficient. Practically very less page faults
2. Doesn't suffer from Be lady's Anomaly

Disadvantages:

1. Complex Implementation
2. Expensive
3. Requires hardware support

Topic : Question

H. W.

- Number of frames = 4
- Pure demand paging used
- Page reference sequence: 5, 7, 0, 1, 7, 6, 7, 2, 1, 6, 7, 6, 1
- Number of page faults for FIFO, optimal and LRU policies?

Topic : Question

H.W.

Consider the following page references:

2, 3, 4, 5, 6, 4, 5, 2, 7, 8, 9, 8, 9, 8, 9, 1, 6, 5, 6, 5, 3

Using optimal policy and 4 frames. Memory access time is 2ms without page fault and 40ms with page fault. The effective memory access time for servicing the above page requests is ____ ms?

2 mins Summary

Topic Page Replacement

Topic FIFO, Optimal, LRU Policies

Happy Learning

THANK - YOU